HW1 EuroSAT RGB MLP Report

Experimental Setup

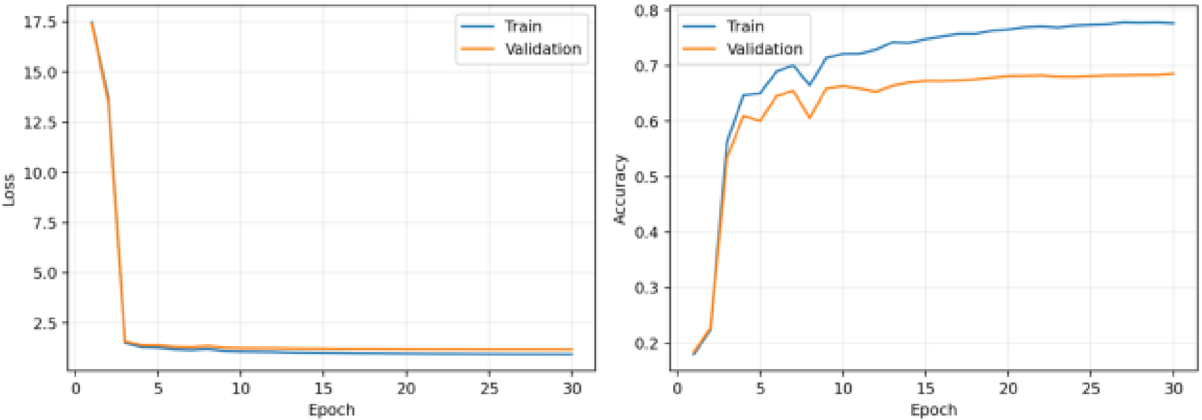
Model: one-hidden-layer MLP, hidden_dim=512, activation=relu.
Input: RGB images resized to 16x16, flattened and standardized.
Split: stratified 70%/15%/15%.
Optimizer: mini-batch SGD, lr=0.16, lr_decay=0.9, batch_size=256.
Loss: cross-entropy + L2 weight decay (0.0005).
Array backend: numpy.

Results

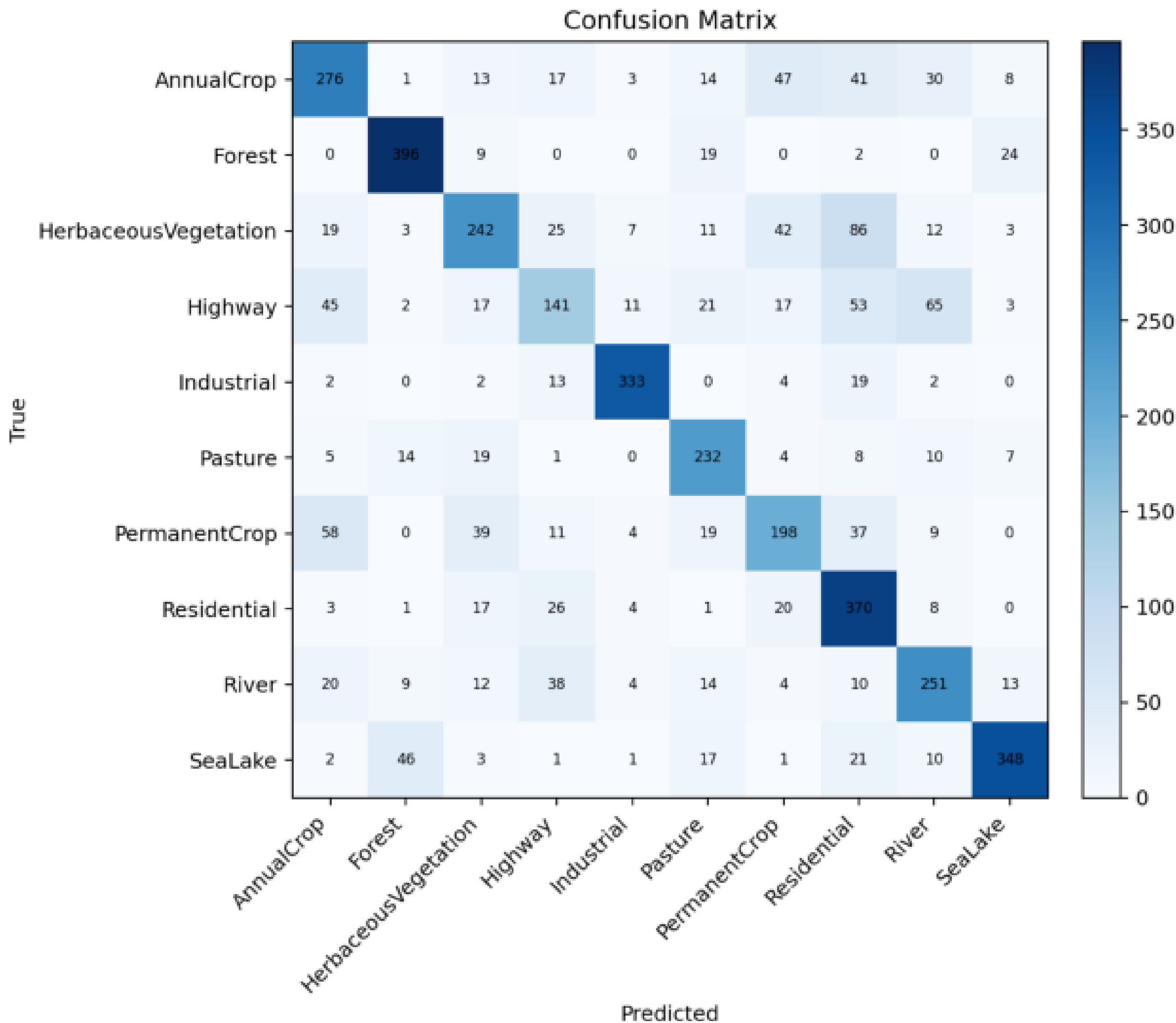
Best validation epoch: 30
Best validation accuracy: 0.6859
Test loss: 1.1440
Test accuracy: 0.6881

Error Analysis

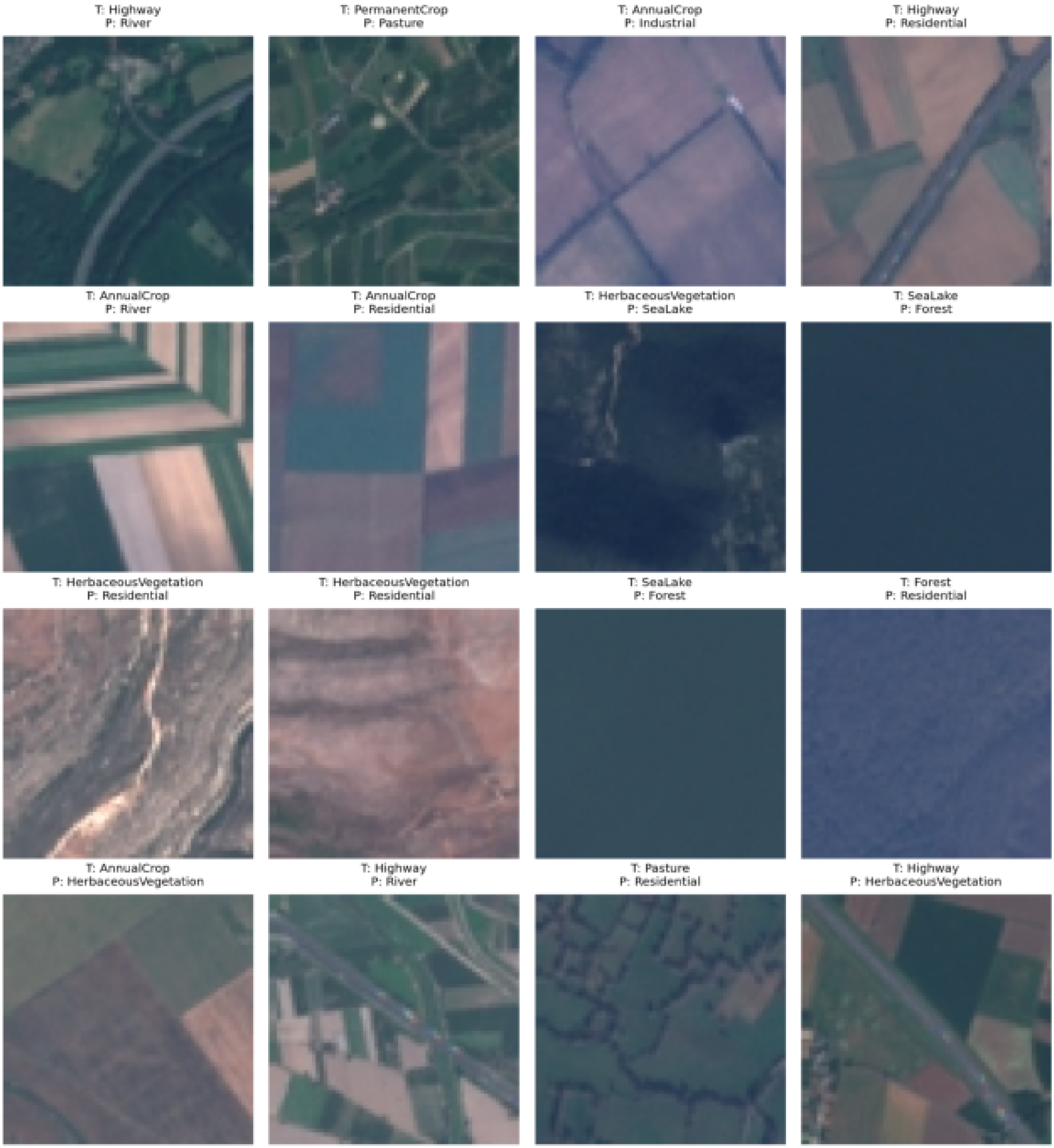
Weakest classes: Highway (0.376), PermanentCrop (0.528), HerbaceousVegetation (0.538).
Likely causes include shared texture and color cues among crop/vegetation classes,
and narrow structures such as River/Highway becoming ambiguous after downsampling.



Train
Validation
Loss
Epoch
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0
5
10
15
20
25
30
Accuracy
0.8
0.7
0.6
0.5
0.4
0.3
0.2



Confusion Matrix
True
Predicted
AnnualCrop
Forest
HerbaceousVegetation
Highway
Industrial
Pasture
PermanentCrop
Residential
River
SeaLake
276 1 13 17 3 14 47 41 30 8
0 396 9 0 0 19 0 2 0 24
19 3 242 25 7 11 42 86 12 3
45 2 17 141 11 21 17 53 65 3
2 0 2 13 333 0 4 19 2 0
5 14 19 1 0 232 4 8 10 7
58 0 39 11 4 19 198 37 9 0
3 1 17 26 4 1 20 370 8 0
20 9 12 38 4 14 4 10 251 13
2 46 3 1 1 17 1 21 10 348
0
50
100
150
200
250
300
350


T: Highway
P: River
T: PermanentCrop
P: Pasture
T: AnnualCrop
P: Industrial
T: Highway
P: Residential
T: AnnualCrop
P: River
T: AnnualCrop
P: Residential
T: HerbaceousVegetation
P: SeaLake
T: SeaLake
P: Forest
T: HerbaceousVegetation
P: Residential
T: HerbaceousVegetation
P: Residential
T: SeaLake
P: Forest
T: Forest
P: Residential
T: AnnualCrop
P: HerbaceousVegetation
T: Highway
P: River
T: Pasture
P: Residential
T: Highway
P: HerbaceousVegetation